Anurag Pradhan

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Portfolio | Google Scholar

Professional Summary

Computer Science Researcher specializing in Deep Learning, Medical AI, and Generative Models. Summer Research Intern at IIT Mandi under Prof. Aditya Nigam, engineering MRI-to-synthetic CT reconstruction pipelines using Diffusion Models, GANs, and Mamba. Proficient in PyTorch, CNNs, Transformers, and VLM architectures. Co-authored an IJCNLP–AACL 2025 publication.

Education

Vellore Institute of Technology | *B.Tech. in Computer Science* **Aug 2023 – Jun 2027** | *Chennai, India*

Sri Venkateswara Jr. College | *Class 12th (MPC)* **Jul 2021 – Jun 2023** | *Visakhapatnam, India*

Experience

Indian Institute of Technology Mandi (IIT Mandi) **May 2026 – Jun 2026**
Research Intern | Deep Learning, Diffusion Models, Mamba, GANs, Vision Encoders *Mandi, Himachal Pradesh*
- Spearheading research on MRI-to-synthetic CT (sCT) reconstruction for radiotherapy planning under Prof. Aditya Nigam, implementing Diffusion Models, GANs, and Mamba architectures.
- Formulating a novel sliding-window DDPM inference technique for high-fidelity 3D volume synthesis, reducing full-volume MAE from 300 HU to 98 HU across 4 anatomical regions on CT scans.

Team Ignition, VIT Chennai **Apr 2024 – Apr 2026**
Flight Software Lead | Arduino, C/C++, PID Control, Image Processing, Deep Learning *On Site*
- Led flight software development across 3 CanSat competition entries, winning all 3.
- Devised fin actuator control, Kalman-filter-based state estimation, PID stability algorithms, and real-time onboard video stabilization, improving telemetry reception reliability by 15% on low-power hardware.

National Aluminium Company Ltd (NALCO) **Jun 2025 – Jul 2025**
Computer Science Intern | Custom YOLOv5, PyTorch, Data Augmentation, Model Evaluation *Damanjodi, Odisha*
- Deployed Respiguard AI, a YOLOv5-based lung disease prediction system for mine worker health screening.
- Fine-tuned on the NIH ChestX-ray14 dataset (112,120 images, 30,805 patients, 14 disease classes); improved detection reliability via targeted augmentation, class-weighted loss, and iterative validation.

Projects

DrishtiAI | *Android, SmolVLM2, llama.cpp, JNI/C++, YOLO-World, Depth-Anything*
- Architected an offline multimodal Android assistant using SmolVLM2 and llama.cpp for on-device visual question answering, enabling real-time local VQA with under 1.8 seconds latency without internet.
- Quantized SmolVLM2-500M to INT8, cutting model size 45% (190MB→104MB) and peak memory 33% (1143MB→762MB) while more than doubling inference throughput (6.41→13.55 tokens/s) on edge hardware.
- Integrated YOLO-World with Depth-Anything for object-aware depth estimation and spatial reasoning.

CMCA-ActionNet – Spatio-Temporal Multimodal Action Detection | *PyTorch, MediaPipe, Optical Flow*
- Engineered a multimodal human action recognition system achieving 84% accuracy on the JHMDB dataset.
- Formulated a Cross-Modal Causal Attention (CMCA) module to fuse RGB, optical flow, and MediaPipe pose landmarks, raising action classification accuracy by 6.2% compared to single-modal baselines.
- Embedded bidirectional temporal attention and an uncertainty-aware prediction head for robust action recognition, cutting overall false positive rates by 15% on unconstrained video sequences.

Brain Tumor Segmentation | *BEiT, Albumentations, PyTorch* **GitHub Repo**
- Designed a BEiT-transformer segmentation pipeline on the BraTS dataset with skip connections and attention-driven decoding, achieving a high 89.2% Dice score for multi-class brain tumors.
- Optimized training via Albumentations and a hybrid CE–Dice loss for stable multi-class convergence.

JanaSathi: Odia E-Governance Chatbot | *RAG, OpenRouter LLMs, Cohere Command-R+* **GitHub Repo**
- Pioneered a bilingual RAG-based chatbot for Odisha government schemes using OpenRouter LLMs and MiniLM embeddings, supporting interactive search over 150+ welfare policies in both English and Odia.
- Deployed Odia translation via Cohere Command-R+; achieved 2nd place in the AMD-sponsored OdiaGenAI Hackathon 2025 by optimizing retrieval precision by 18% using hybrid search algorithms.

Conference Publications

Towards Blind and Low-Vision Accessibility **Paper Link**
- *of Lightweight VLMs and Custom LLM-Evals* – S. S. Baghel, Y. P. S. Rathore, A. Pradhan. Accepted at the 14th International Joint Conference on Natural Language Processing & the 4th Asian Chapter of ACL (IJCNLP–AACL 2025) in Mumbai, India, which has an acceptance rate of ∼22%.

Technical Skills

Programming Languages: Python, C, C++, Java, Kotlin
Core Expertise & Research: VLMs, LLMs, EdgeAI, Agentic/Generative AI, Action Recognition, Quantization
Frameworks & Libraries: PyTorch, TensorFlow, Transformers, llama.cpp, BEiT, BLIP, OpenCV, LangChain
Tools & Platforms: Git, Docker, CUDA, ONNX, Weights & Biases, Hugging Face Hub, Android/Compose (JNI)